

Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds

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Abstract

Homeostatic property cluster theory gave natural-kinds realism a powerful anti-essentialist form: a kind can be real because its properties cluster through mechanisms rather than because it has an essence (Boyd, 1991, 1999). But the word HOMEOSTATIC has often carried more weight than the theory has earned. It can name reliable co-occurrence, internal causal order, or feedback-sensitive control. These are different explanatory achievements.

This paper treats projectibility as the target and offers a diagnostic for how it is secured. Stable property clusters require reliable co-occurrence with a projectible payoff. Networked kinds require internal structure among central, derivative, and evidentially useful properties. Control-secured kinds require a stronger pattern: perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation. Weinberger's account of homeostasis as causal control supplies the guardrail (Weinberger, 2026).

The Watt governor is the calibration case, glycemic regulation is a harder biological extension, and English interjections are a stable pragmatic contrast case. The result is a disciplined way to use Boyd's insight without turning HOMEOSTATIC into a loose synonym for stable, mechanism-maintained, or projectible.

I INTRODUCTION

The target is a quiet ambiguity in recent anti-essentialist kind theory. Boyd's homeostatic property cluster framework, Slater's stable property cluster proposal, and Khalidi's causal-network account all reject essences. They also all aim to explain why some categories support projection. The ambiguity concerns what secures that projection: clustering, observed stability, internal order, or homeostatic maintenance (Boyd, 1991, 1999; Khalidi, 2013, 2018; Slater, 2015).

The working claim is simple: not every stable cluster is homeostatic. Treating it as such makes the framework too permissive. The correction keeps Boyd's projectibility frame while making HOMEOSTASIS more literal and more demanding. Weinberger's causal-control account supplies the guardrail (Weinberger, 2026).

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The contribution is diagnostic. The paper does not reject HPC theory; it separates three explanatory roles that HPC discussions often compress. First, some profiles support projection because their properties recur reliably. Second, some profiles support projection because their properties occupy ordered positions in a causal or mixed network. Third, some profiles support projection because feedback-sensitive organization corrects perturbation and preserves a higher-scale relation. Only the third role earns the word HOMEOSTATIC.

2 PROJECTIBILITY FIRST

Boyd's framework works because projectibility does not require necessary and sufficient conditions. A category can support induction when its associated properties are not accidental, even if no property is shared by every member. Boyd's accommodation thesis ties successful classification to the way our inferential practices fit causal structure, not to definitions fixed in advance (Boyd, 1991, 1999).

This point is easy to weaken. If the only question is whether a cluster recurs, then many categories will qualify too cheaply. Stable co-occurrence may support some inference, but the paper still has to say what can be projected from what. A stable category becomes theoretically interesting when its profile licenses prediction, explanation, or controlled intervention.

3 OBSERVED STABILITY AND INVARIANCE

Slater's stable property cluster account usefully presses the point that homeostatic mechanisms are not always needed for natural kindness (Slater, 2015). Some categories support induction because their properties form a robust cluster, even when no corrective mechanism returns the system to a set point. That is the first distinction this paper needs: a stable projectible cluster may be real without being homeostatic.

Stability is also too thin if it is left unexplained. Onishi and Serpico argue that stable-cluster views risk losing the processes that make some inductions rational, while strict mechanism views inherit boundary, abstraction, and one-to-many problems (Onishi & Serpico, 2022). Woodward's account of invariance helps sharpen the issue: explanatory stability holds under relevant changes and interventions, not just across space and time (Woodward, 2001).

The first diagnostic is therefore modest. A stable property cluster should support some projection under a specified range of contexts. The evidence cannot be mere recurrence. It has to show that observing part of the profile licenses a useful expectation about another part, or about the behaviour of the system in a relevant setting.

4 NETWORK ORDER WITHOUT CONTROL

Khalidi's network account addresses a different weakness. A cluster is more than a list of associated properties. Its properties can be ordered: some generate, sustain, constrain, or explain others. That is an advance over a bare cluster picture because it explains why some properties are diagnostic and others derivative (Khalidi, 2013, 2018).

Still, ordered dependence falls short of corrective control. A node in a causal network may support projection because nearby properties stand in systematic causal relations. That does not mean the system tracks departures, allows lower-scale variables to adjust, and preserves a higher-scale relation through that adjustment. Network order can explain projectibility without being homeostasis.

The second diagnostic asks whether the profile has internal structure. Relevant evidence includes asymmetries among properties: upstream features, derivative features, central features, and peripheral cues. The failure mode is causal vocabulary used for an unordered list. A paper does not establish network order merely by redescribing co-occurrence as causal.

I use NETWORK ORDER broadly in the cross-domain diagnostic. In strictly biological or mechanical cases, the ordering may be causal. In linguistic and social cases, the relevant dependencies may also be functional, conventional, sequential, or evidential. What matters is that the profile has structure that explains projection. A merely associated bundle of features is not enough.

5 CORRECTIVE CONTROL

Homeostatic-control-secured kinds are stricter. Kinds do not themselves sense, compare, or correct. A kind earns the label only when its projectible profile is maintained by a system or practice with feedback-sensitive corrective organization. Weinberger's central lesson is that homeostasis requires more than constancy. A controlled system may preserve a higher-scale relation through lower-scale variation (Weinberger, 2026).

For this paper, corrective control has four evidential marks. First, a perturbation threatens the projectible profile. Second, a tracked variable or relation matters to the profile. Third, a response pathway changes lower-scale conditions. Fourth, that response preserves a higher-scale relation that supports projection. Feedback alone is not enough; the feedback has to explain why the profile remains projectible under perturbation.

The biological control literature makes the same warning concrete. Control mechanisms sense conditions, transmit signals, and alter constraints in controlled mechanisms (Bich & Bechtel, 2022a). In organisms, such controls are integrated rather than centrally commanded (Bich & Bechtel, 2022b; Bich et al., 2016). That is much more specific than saying that a property cluster is stable, or that its properties are causally ordered.

6 THREE DIAGNOSTICS

The proposal is best understood as a diagnostic grid, not as a simple ladder. Homeostatic-control-secured cases will usually display stability and network order, but the reverse inference fails. A stable cluster need not have an internally ordered causal profile, and a network-ordered kind need not be maintained by corrective control.

The grid turns the question around. For any proposed HPC kind, the diagnostic asks three questions before using HOMEOSTATIC. First, what does the category let us project? Second, what structure secures that projection: stable co-occurrence, network order, or corrective control? Third, what evidence would move the case from one diagnostic to another?

7 HOW TO READ THE CASES

The cases below do not all have the same status. The Watt governor is not offered as a kind. It is a calibration case for the control structure that a homeostatic-control-secured kind would have to involve. Glycemic regulation is a candidate biological domain where that calibration can be applied. Interjections are a linguistic category whose stable projectibility is clear enough to test whether stability and interactional order license the stronger homeostatic label.

Claim being made	Evidence needed	Failure mode
Stable cluster	Reliable co-occurrence across relevant contexts, plus a specified projection from observed to unobserved features	Mere recurrence without projectible payoff
Network-ordered kind	Internal ordering among properties, with some properties upstream, central, derivative, or evidentially privileged	Causal vocabulary used for an unordered list
Homeostatic-control-secured kind	Perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation	Treating stability, maintenance, or feedback as control

Table 1: Diagnostics for three ways projectibility can be secured.

8 CALIBRATION CASE: WATT GOVERNOR

Weinberger's Watt governor is the calibration case because the control structure is explicit. Changes in engine speed change the governor's flyball position; that adjustment changes valve aperture; valve aperture changes steam flow. The point is not that shaft speed is clamped. A clamp intervention on shaft speed would sever the incoming causal relations and destroy the control relation (Weinberger, 2026).

The governor illustrates why HOMEOSTATIC should not mean constant. Its lower-scale variables are allowed to vary so that workload and steam flow remain coupled in a higher-scale relation. This is the strict pattern the diagnostic is looking for: perturbation, responsive lower-scale adjustment, and preservation of a relation that supports projection.

9 BIOLOGICAL EXTENSION: GLYCEMIC REGULATION

Glycemic regulation is useful because it is harder than the governor. The relevant candidate is not glucose as a substance, nor a single blood value, but the normoglycemic regulatory profile of an organism. Blood-glucose regulation involves sensors, signals, effectors, and organized adjustment across feeding, fasting, activity, and other perturbations (Bich et al., 2020). It therefore looks like a natural biological domain in which a projectible profile might be control-secured.

The case also keeps the diagnostic away from ordinary SET POINT talk. Population correlations for glucose and calcium may fit an equilibrium model better than a defended set point (Fitzgerald & Bean, 2018). That challenge does not make glycemic regulation irrelevant. It makes the evidential question more precise: does regulation merely settle into a stable range, or does it preserve a higher-scale relation through organized lower-scale variation?

Bich et al. (2020) answer by moving beyond a simple feedback-loop diagram. Glycemia regulation sits inside wider organizational closure, where multiple constraints and processes make glucose available to the organism while keeping concentrations within a viable range. The lesson for kind theory is direct. A biological example counts as homeostatic-control-secured only when the evidence shows this organized correction, not when the variable is merely stable.

10 CONTRAST CASE: INTERJECTIONS

English interjections are the negative linguistic case. They appear stable and projectible in use: prosodic isolation, low syntactic integration, non-inflection, non-referentiality, and interactional force often travel together (Ameka, 1992; Norrick, 2009). Observing some of these properties licenses expectations about others. That is a projectible profile, not a mere list.

The projectible payoff is pragmatic. Forms such as *huh?*, *mhm*, and *oh* support expectations about sequential position and interactional uptake: repair initiation, continuation, and change of epistemic state (Dingemanse, 2024). Forms such as *wow*, *oops*, *hey*, *well*, and *uh* invite expectations about stance, event management, attention, transition, or delay. Norrick (2009) is useful here because he treats interjections as an open-ended class with recurrent pragmatic-marker functions.

The list is deliberately heterogeneous. Some of these forms are analysed as discourse markers, fillers, summonses, or response tokens rather than as central interjections. That is part of the test. The contrast case is not a classical kind with settled boundaries; it is a projectible interactional profile whose kindhood is itself partly at stake.

The likely stabilizers are conventionalization, turn-taking routines, prosodic recognizability, acquisition, register, uptake, and metapragmatic learning. These make the category stable and useful. They may also give parts of the profile network order: sequential position, prosodic packaging, and syntactic non-integration are not independent features.

The homeostasis question is stricter. For interjections, the higher-scale relation would be an interactional profile linking turn position, prosodic packaging, low integration, and uptake. Lower-scale variables would include token form, intonation, sequential placement, phonological reduction, and local conventional meaning. Perturbations would include syntactic integration, inflection, shifted prosody, misplaced sequential use, or uptake failure.

A homeostatic-control claim would require evidence that such perturbations trigger coordinated correction that preserves the higher-scale interactional relation. Repair, uptake failure, teaching, and metalinguistic policing may sometimes provide control-like evidence. But that evidence has to be shown. Without it, interjections belong in the stable or network-ordered diagnostics. They are a useful case because they make the temptation visible and then block the overclaim.

11 DISCUSSION

The distinction matters for the broader HPC programme in linguistics and social ontology. Linguistic categories may be stable, projectible, and mechanism-maintained without all being homeostatic in the strict control-theoretic sense. Social practices may be stabilized by records, offices, sanctions, training, and uptake without becoming single homeostatic control systems.

A natural objection is that social correction and metalinguistic policing are feedback. Sometimes they are. That concession strengthens the diagnostic rather than weakening it. It means some linguistic or social categories may be homeostatic-control-secured, but only where correction, repair, acquisition, transmission, and uptake form a coordinated control architecture. Grammaticality or a contested social category may be a better positive linguistic case than interjections. The present point is that the evidence has to earn that upgrade.

Grammaticality shows what a positive linguistic case would require. The higher-scale relation would be a community-sensitive pattern linking forms, judgments, parsing expectations, correction, and transmission. Lower-scale variables would include individual productions, acquisition episodes,

classroom correction, repair sequences, editorial practices, and analogical levelling. A homeostatic-control claim would need evidence that departures from the profile reliably elicit coordinated responses that preserve the projectible relation across speakers and contexts. The diagnostic leaves that possibility open, but it does not grant it by default.

12 CONCLUSION

Projectibility remains the point. Mechanisms and network order matter because they explain why projection is reliable. Homeostasis is one important way projection can be secured, but the term should be reserved for corrective control. The diagnostic protects Boyd's anti-essentialist insight while preventing HOMEOSTATIC from becoming a loose synonym for stable, useful, or causally organized.

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REFERENCES

- Ameka, F. (1992). Interjections: The universal yet neglected part of speech. *Journal of Pragmatics*, 18(2–3), 101–118. [https://doi.org/10.1016/0378-2166\(92\)90048-G](https://doi.org/10.1016/0378-2166(92)90048-G)
- Bich, L., & Bechtel, W. (2022a). Control mechanisms: Explaining the integration and versatility of biological organisms. *Adaptive Behavior*, 30(5), 389–407. <https://doi.org/10.1177/10597123221074429>
- Bich, L., & Bechtel, W. (2022b). Organization needs organization: Understanding integrated control in living organisms. *Studies in History and Philosophy of Science*, 93, 96–106. <https://doi.org/10.1016/j.shpsa.2022.03.005>
- Bich, L., Mossio, M., Ruiz-Mirazo, K., & Moreno, A. (2016). Biological regulation: Controlling the system from within. *Biology & Philosophy*, 31(2), 237–265. <https://doi.org/10.1007/s10539-015-9497-8>
- Bich, L., Mossio, M., & Soto, A. M. (2020). Glycemia regulation: From feedback loops to organizational closure. *Frontiers in Physiology*, 11, Article 69. <https://doi.org/10.3389/fphys.2020.00069>
- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61(1–2), 127–148. <https://doi.org/10.1007/BF00385837>
- Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. <https://doi.org/10.7551/mitpress/6396.003.0012>
- Dingemanse, M. (2024). Interjections at the heart of language. *Annual Review of Linguistics*, 10(1), 257–277. <https://doi.org/10.1146/annurev-linguistics-031422-124743>
- Fitzgerald, S. P., & Bean, N. G. (2018). Population correlations do not support the existence of set points for blood levels of calcium or glucose: A new model for homeostasis. *Physiological Reports*, 6(1), Article e13551. <https://doi.org/10.14814/phy2.13551>
- Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511998553>

- Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, 195(4), 1379–1396. <https://doi.org/10.1007/s11229-015-0841-y>
- Norricks, N. R. (2009). Interjections as pragmatic markers. *Journal of Pragmatics*, 41(5), 866–891. <https://doi.org/10.1016/j.pragma.2008.08.005>
- Onishi, Y., & Serpico, D. (2022). Homeostatic property cluster theory without homeostatic mechanisms: Two recent attempts and their costs. *Journal for General Philosophy of Science*, 53, 61–82. <https://doi.org/10.1007/s10838-020-09527-1>
- Slater, M. H. (2015). Natural kindness. *The British Journal for the Philosophy of Science*, 66(2), 375–411. <https://doi.org/10.1093/bjps/axt033>
- Weinberger, N. (2026). Homeostasis and causal control. *Biology & Philosophy*, 41(2), Article 18. <https://doi.org/10.1007/s10539-026-10018-8>
- Woodward, J. (2001). Law and explanation in Biology: Invariance is the kind of stability that matters. *Philosophy of Science*, 68(1), 1–20. <https://doi.org/10.1086/392863>